02-deuring-lift: canonical CM(−19) lift of the pair-46 crater

Pair-46 factor-base campaign, item 02 · 2026-10-15 · protocol `PROTOCOL.md` (frozen before data) · data `results.json`, `summary.json`, `run.log`.

Verdict: exact obstruction (prediction "fails by rank obstruction" confirmed).

Exact statements

1. The Deuring lift of the crater is Cremona **361a1**, $y^2 + y = x^3 - 38x + 90$, with $j = -96^3$ and conductor 361. Its a_p is $t = 34489701$, so $\#E(\mathbb{F}_p) = N$, and the explicit isomorphism to the crater (with u^2 a square mod p) is verified. No twist is needed.
2. $E(\mathbb{Q}) = \mathbb{Z}G$ with $G = (0, 9)$. The rank is 1 unconditionally (ellrank bounds [1,1]; analytic rank 1), torsion is trivial, and $\hat{h}(G) = 0.181292$.
3. $G \bmod p$ has order N, so reduction is onto. The lift of a target $T = kG$ is kG, with height $k^2\hat{h}(G)$, and its expected height is $\approx 2.1 \cdot 10^{30}$ nats. Lifting is therefore the DLP itself, and the only relations that involve the target are the trivial ones once k is known.
4. Twists E_d ($1 < |d| \le 200$, one per $\{d, -19d\}$ class, 115 curves): 51 have rank 0, 54 rank 1, 8 rank 2 and 2 rank 3. Generators on distinct twists are $\mathbb{Z}$-independent (they lie in distinct Galois eigenspaces), so a twist factor base has **0** $\mathbb{Q}$-relations. Every relation among them exists only mod p, as a point of a determinant-N kernel lattice.
5. Over $\mathbb{Q}(\sqrt{-19})$ the only extra relation is $wG \equiv \lambda G$, the known GLV relation, which the glv-orbit item already covers. This one is a derivation and was not computed here.

Empirical (24 toy primes at 16/20/24 bits, $R = 6$)

metric	A2 canonical	NC random	PC planted
λ_1/ GH median	1.393	1.250	0.290 (max 0.461)
A2 − NC (95% CI)	+0.143 [−0.180, +0.262], Mann–Whitney p = 0.50		
coverage diff	+0.0008 [−0.052, +0.056]		
lifted Q-relations	0 / 24 primes	n/a	48 (planted 2 per prime)
Xedni generic $r = 2..5$	0 / 96 trials	n/a	24 / 24 ($P, -P$ planted)

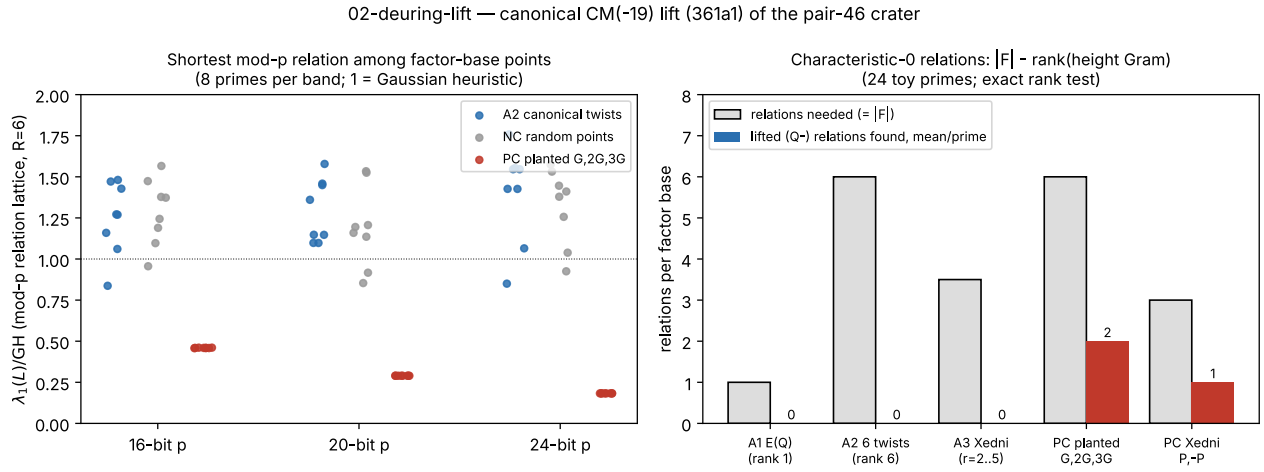


Figure 1: Left: shortest mod-p relation among $R = 6$ factor-base points, normalized by the Gaussian heuristic. The canonical-lift generators (A2) cannot be distinguished from random points (NC), while the planted dependency (PC) is caught at every prime. Right: characteristic-0 relations (|F| − rank of the canonical-height Gram matrix) against the number a factor base needs. 24 toy primes; run `run_deuring_lift.py`, seed 20261015.

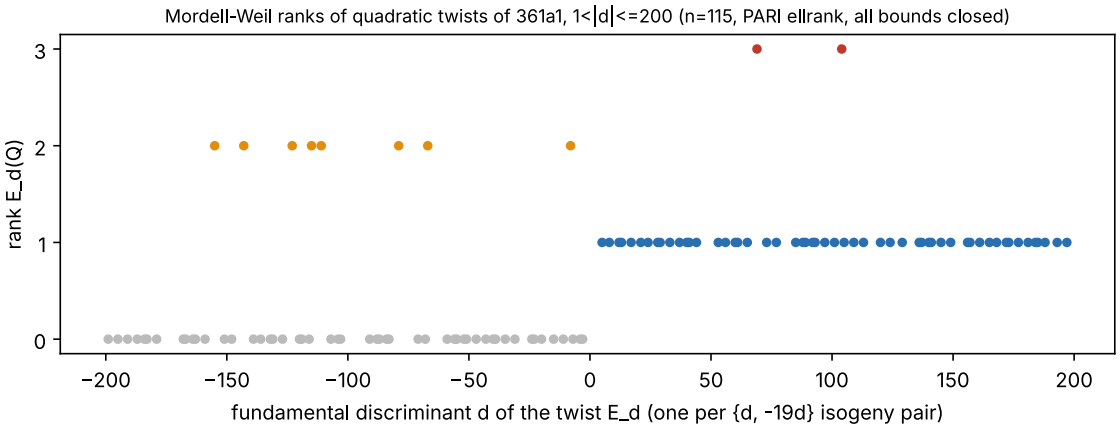


Figure 2: Mordell–Weil ranks of the quadratic twists of 361a1 for $1 < |d| \leq 200$ (PARI ellrank, every bound closed). The largest rank is 3.

Deviations and scope

- **Precision fix.** cypari2 defaults to 15 digits of precision. At that precision the positive control (the planted $G, 2G, 3G$ Gram matrix) failed the frozen 10^{-20} tolerance, so I raised realprecision to 80 for every arm. The tolerance itself was not changed.
- **Nothing cut.** Nothing in the preregistered design was cut, and the whole run took 2.4 s on 1 core.
- **Limits.** One generator per twist, $|d| \leq 200$, and no DLP at pair-46 size. The pair-46 statements are exact.

Reproduce: `python3 run_deuring_lift.py results.json && python3 analyze.py && python3 -c "import typst; typst.compile('report.typ', output='report.pdf')"`